

Foundations of Research Methods

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SESSION 1 – RESEARCH CONCEPT

Learning Outcomes

- At the end of the session, students should be able to:
- Define research in the context of IT and system development
- Explain the significance of research in web, mobile, and analytics projects
- Identify the characteristics of good research
- Describe the purposes of research in capstone projects

What is research?

Research is a **systematic and organized investigation** conducted to:

- answer questions,
- solve problems, and
- improve existing systems or processes.

In Information Technology, **research is not purely theoretical**. It is **applied**, meaning:

- You research **before** building a system
- You research **while** developing the system
- You research **after** deployment to evaluate effectiveness

For BSIT students:

- A **web system, mobile app, or analytics dashboard** must be justified by research.
- You do not simply “code an idea.”
- You **study a problem**, gather data, analyze it, and then design a solution.

Your capstone project is a research-based system, not just a software project.

Example

- A library system is not built just because “it is needed.”
- Research determines:
 - What problems exist?
 - Who experiences them?
 - What features are necessary?
 - How effective is the system after implementation?

Significance of Research

Research is important because it:

1. Solves real-world problems

- Delays, inefficiencies, inaccurate data, poor services

2. Supports data-driven decision-making

- Especially important in **analytics-based systems**

3. Improves system quality

- Usability, performance, security, reliability

4. Justifies system development

- Prevents unnecessary or irrelevant systems

5. Contributes to institutional improvement

- Universities, LGUs, businesses

Connection

- Web systems → improve accessibility
- Mobile apps → improve convenience
- Analytics → improve **planning, forecasting, and decision-making**

Without research:

- Systems become underutilized
- Clients reject the solution
- Capstone projects fail evaluation

Characteristics of Research

A good research study has the following characteristics:

Systematic

- . Follows a step-by-step process
- . Similar to **SDLC** (System Development Life Cycle)

Objective

- . Based on facts and data, not opinions
- . Avoids bias

Empirical

- . Based on observable and measurable evidence
- . Surveys, logs, system usage data

Analytical

- Data are examined and interpreted
- Especially important for analytics specialization

Replicable

- Other researchers can repeat the study
- Results can be verified

Purpose of Research

Research is conducted to:

1. Describe

- Current systems, workflows, or problems

2. Explain

- Why problems occur
- Relationships between variables

3. Predict

- Future outcomes using analytics
- Trends, demand, performance

4. Improve

- Services, processes, and systems

5. Innovate

- Introduce smarter, more efficient solutions

Analytics Emphasis

For students specializing in **Data & Business Analytics**:

- Research enables:
 - Predictive analytics
 - Decision support systems
 - Performance optimization

SESSION 2 – CATEGORIES OF RESEARCH

Learning Outcomes

- At the end of this session, students should be able to:
- Differentiate qualitative and quantitative research
- Identify appropriate research types for capstone projects
- Explain common research methods used in BSIT
- Select suitable research approaches for web, mobile, and analytics systems

Categories of Research

Research can be classified based on:

- the **nature of data**, and
- the **purpose of the study**.

Understanding research categories is important because:

- it determines **how data will be collected**,
- it affects **system evaluation**, and
- it guides **analytics implementation**.

Qualitative Research

Qualitative research focuses on:

- meanings,
- experiences,
- perceptions, and
- behaviors.

It answers “**WHY**” and “**HOW**” questions.

Common Qualitative Data Sources

- Interviews
- Focus group discussions
- Observations
- Open-ended questions
- Documents and reports

Characteristics

- Data are **non-numerical**
- Results are **descriptive**
- Small sample sizes
- In-depth analysis

Qualitative research is useful when:

- understanding **user experience (UX)**,
- identifying **user problems and expectations**, or
- gathering feedback before system development.

Example uses

- Interviewing librarians about workflow issues
- Gathering student feedback on mobile app usability

Quantitative Research

Quantitative research focuses on:

- numerical data,
- measurement, and
- statistical analysis.

It answers “**HOW MANY**”, “**HOW OFTEN**”, and “**HOW MUCH**” questions.

Common Quantitative Data Sources

- Surveys with Likert scales
- System usage logs
- Performance metrics
- Test results
- Analytics dashboards

Characteristics

- Data are numerical
- Uses statistical tools
- Larger sample sizes
- Results are generalizable

Most **capstone projects are quantitative** because:

- system quality must be **measured**,
- usability, performance, and reliability must be evaluated,
- analytics requires numerical data.

Examples

- User satisfaction ratings
- System response time
- Accuracy of analytics predictions

Qualitative vs. Quantitative Research (Comparison)

Aspect	Qualitative	Quantitative
Data Type	Textual	Numerical
Purpose	Understand behavior	Measure performance
Tools	Interviews, observations	Surveys, statistics
Sample Size	Small	Medium to large
Use in Capstone	Supporting	Primary

Most capstone projects use **quantitative research with some qualitative support.**

Types of Research

a. Descriptive Research

- . Describes the current situation or system

Application:

- . Describing existing manual or computerized systems
- . Presenting current problems

Example:

- . Describing how a university currently manages library transactions

b. Applied Research

Purpose:

- Solves practical, real-world problems

Application:

- Designing systems for universities, LGUs, or businesses

Most capstone projects fall under applied research.

Developmental Research

- Focuses on system or product development

Application:

- Web systems
- Mobile applications
- Analytics dashboards

Capstone projects are also developmental research.

d. Action Research

- Improves existing practices or processes

Application:

- Improving an existing system
- Enhancing workflows

Common in institutional settings

Research Methods

Research methods describe **HOW the study is conducted.**

a. Survey Method

- Most common in BSIT capstone
- Used to gather user feedback
- Often uses Likert scale questionnaires

Used for:

- usability testing
- system evaluation
- ISO/IEC 25010 assessment

b. System Development Life Cycle (SDLC)

- . Planning
- . Analysis
- . Design
- . Development
- . Testing
- . Deployment
- . Maintenance

SDLC is both:

- . a development framework, and part of the research methodology

Case Study Method

- Focuses on one organization or client
- Common in LGU- or university-based projects

Data Analysis and Analytics Method

- Descriptive analytics
- Diagnostic analytics
- Predictive analytics

Used by students with **Data & Business Analytics specialization**

SESSION 3 – THE RESEARCH PROBLEM

Learning Outcomes

- At the end of this session, students should be able to:
- Define what a research problem is
- Identify the elements of a research problem
- Describe the characteristics of a good research problem
- Identify sources of research problems
- Apply criteria in selecting a feasible capstone research problem

Research Problem

A **research problem** is a **clear statement of an issue, gap, or difficulty** that:

- exists in a real-world setting,
- affects specific users or stakeholders, and
- requires investigation and a solution.

In BSIT capstone projects:

- The research problem **justifies the need for the system**.
- No clear problem → no valid capstone project.

A system without a research problem is just a software demo.

Examples of research problems

- Inefficient manual processes
- Data inaccuracy and redundancy
- Lack of real-time reporting
- Poor accessibility or usability
- Absence of analytics for decision-making

Importance of a Well-Defined Research Problem

A well-defined research problem:

- guides the **objectives of the study**,
- determines the **scope and features of the system**,
- influences the **research design and methodology**,
- ensures the study is **relevant and feasible**.

Without a clear problem:

- objectives become vague,
- systems become too broad,
- evaluation becomes meaningless.

Elements of a Research Problem

A complete research problem contains the following elements:

a. The Issue or Gap

What is wrong?

- Inefficiency?
- Inaccuracy?
- Delay?
- Lack of automation?
- Poor decision support?

b. Target Users or Clients

Who is affected?

- Students
- Faculty
- Administrators
- LGU personnel
- Business staff

c. Variables or Data Involved

What data are related to the problem?

- Records
- Transactions
- Reports
- Analytics data

d. Context or Environment

Where does the problem exist?

- State universities
- LGUs
- Small businesses
- Specific departments or offices

A research problem must clearly answer **WHAT, WHO, and WHERE.**

Characteristics of a Good Research Problem

A good research problem must be:

Clear and Specific

- . Clearly stated
- . Avoids broad or vague terms

Bad: “The system is inefficient”

Good: “Manual record processing causes delays in report generation”

Researchable

- Data can be collected
- Methods exist to investigate the problem

Measurable

- Can be evaluated using metrics
- Important for system evaluation (e.g., ISO/IEC 25010)

Relevant

- Addresses real institutional needs
- Useful to clients and stakeholders

Feasible

- Can be completed within:
 - time constraints,
 - available resources,
 - student skill level.

Ethical

- Does not violate privacy or ethical standards

Sources of Research Problems

Students may derive research problems from:

a. Personal Observation

- . Internship experiences
- . School or community involvement

b. Existing Systems

- . Manual or outdated systems
- . Systems with limited features

c. Literature Review

- . Gaps identified in previous studies

d. Institutional Needs

- . Universities
- . LGUs
- . Offices requesting automation

e. Emerging Trends

- . Digital transformation
- . Analytics and decision support systems
- . Mobile-first solutions

Criteria for Selecting a Research Problem

Before finalizing a capstone topic, students must evaluate the problem using the following criteria:

a. Interest of the Researchers

- . Motivation is necessary for project completion

b. Availability of Data

- . Data must be accessible and sufficient

c. Technical Feasibility

- . Skills in web, mobile, and analytics must match project requirements

d. Time and Resource Constraints

- . Must fit within the academic calendar

e. Client Support

- . Presence of a willing client or institution

f. Alignment with Program Outcomes

- . Must align with BSIT specialization

SESSION 4 – SAMPLING PROCEDURE

Learning Outcomes

At the end of the session, students should be able to:

- Define population and sampling
- Explain the importance of sampling in research
- Differentiate probability and non-probability sampling
- Identify appropriate sampling techniques for capstone projects
- Apply sampling procedures in system evaluation studies

Population and Sampling

Population

The **population** refers to the **entire group of individuals** relevant to the study.

Examples:

- All students of a state university
- All employees of an LGU office
- All users of a specific system

Sampling

Sampling is the process of selecting a **subset of the population** to represent the whole.

Studying the entire population is often impractical; sampling makes research feasible.

Importance of Sampling in Research

Sampling is important because it:

- saves time and resources,
- allows data collection within project constraints,
- enables system evaluation,
- supports statistical analysis,
- ensures valid and reliable conclusions.

For capstone projects:

- Sampling is commonly used during **system testing and evaluation**.
- Respondents usually include **users, staff, or administrators**.

Types of Sampling Procedures

Sampling procedures are classified into:

1. Probability Sampling

2. Non-Probability Sampling

Probability Sampling Procedures

Probability sampling is a method where **every member of the population has an equal chance of selection.**

Used when:

- population size is known,
- randomness is required,
- generalization is important.

a. Simple Random Sampling

- Every member has an equal chance
- Uses random selection

Example:

- Randomly selecting students from an enrollment list

b. Systematic Sampling

- Selection follows a fixed interval (e.g., every 5th respondent)

Example:

- Selecting every 10th student entering the library

c. Stratified Sampling

- Population divided into groups (strata)
- Samples taken from each group

Example:

- Students grouped by year level

d. Cluster Sampling

- Population divided into clusters
- Entire clusters are selected

Example:

- Selecting entire classes or departments

Limitation in Capstone

- Often difficult due to access and time constraints

Non-Probability Sampling Procedures

Non-probability sampling is a method where **not all members have an equal chance of selection.**

Most capstone projects use non-probability sampling.

a. Convenience Sampling

- Selection based on availability

Example:

- Students who are easily accessible

Purposive Sampling

- Selection based on specific criteria

Example:

- Selecting librarians or system administrators

c. Quota Sampling

- Fixed number of respondents per group

Example:

- 20 students, 10 faculty members

d. Snowball Sampling

- Existing respondents recommend others

Example:

- Useful when respondents are difficult to identify

Choosing the Appropriate Sampling Method

Consider the following:

1. Population size
2. Accessibility of respondents
3. Purpose of the study
4. Time and resource limitations
5. Research design

Sampling method must be clearly justified in Chapter III.

Sampling in System Evaluation

Sampling is commonly used to:

- evaluate usability,
- assess performance,
- measure user satisfaction,
- validate analytics accuracy.

Example:

- Surveying system users using ISO/IEC 25010 criteria

Common Sampling Errors

Sampling Bias

- Selecting only favorable respondents

Small Sample Size

- Results may not be reliable

Unrepresentative Sample

- Results do not reflect the population

Solution:

- Clearly justify sampling choice and limitations

Thank you